Domain Aware C MAPSS Predictive Maintenance

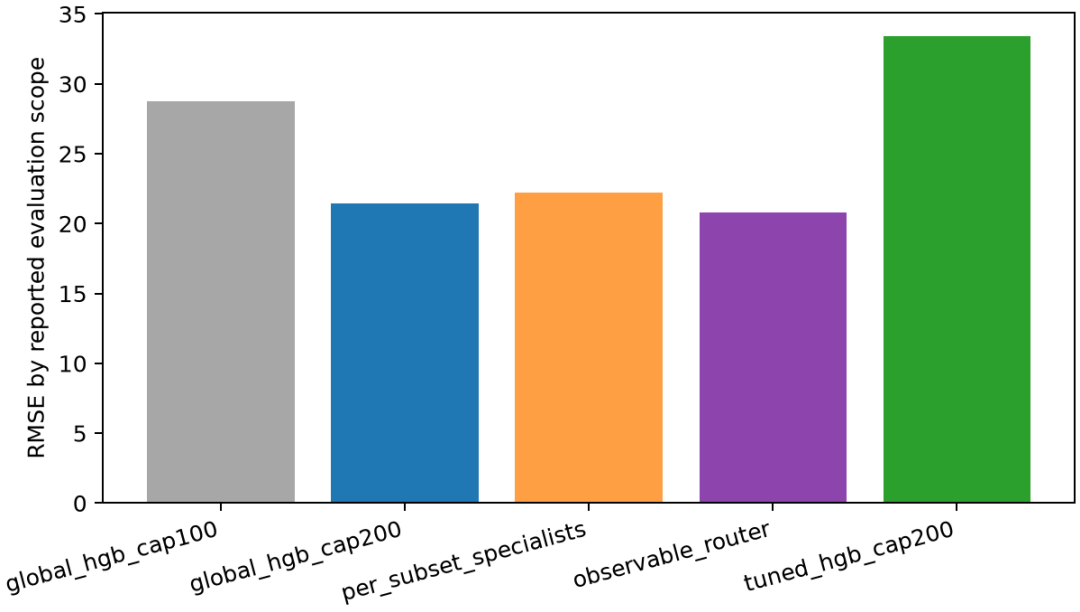
TELIPORT Case Study Final Measured Report

Purpose: document the audit, experiments, operational prototype and evidence-based recommendation for an OPP presentation.

Executive conclusion

The recommended benchmark candidate is a global condition-normalized HGB model with a 200-cycle training cap. It lowers the pooled historical official-test RMSE from 28.737 to 21.455 cycles, while the evidence also records its FD001 and FD003 regression. All official test scores are a historical audit because those labels had already been inspected before this study.

Model	RMSE	MAE	R2	PHM08	Decision
global_hgb_cap100	28.737	18.006	0.683	45.454	cap200 was validation-selected; cap100 benchmark
global_hgb_cap200	21.455	14.622	0.824	31.465	cap200 was validation-selected; cap100 benchmark
per_subset_specialists	22.206	15.009	0.811	37.539	rejected worse pooled result
observable_router	20.797	14.275	0.834	26.159	diagnostic only: failed validation gate
tuned_hgb_cap200	33.401	20.419	0.758	292864.996	RMSE challenger; PHM08 gate failed



Audit and evaluation protocol

All 8 C-MAPSS train/test files passed schema, completeness, numerical-integrity, engine-cycle uniqueness and alignment checks. Development uses engine-grouped validation and six causal pseudo-censored endpoints per held-out engine: 15%, 30%, 45%, 60%, 75% and 90% observed life. Every fitted transform is learned only on development engines.

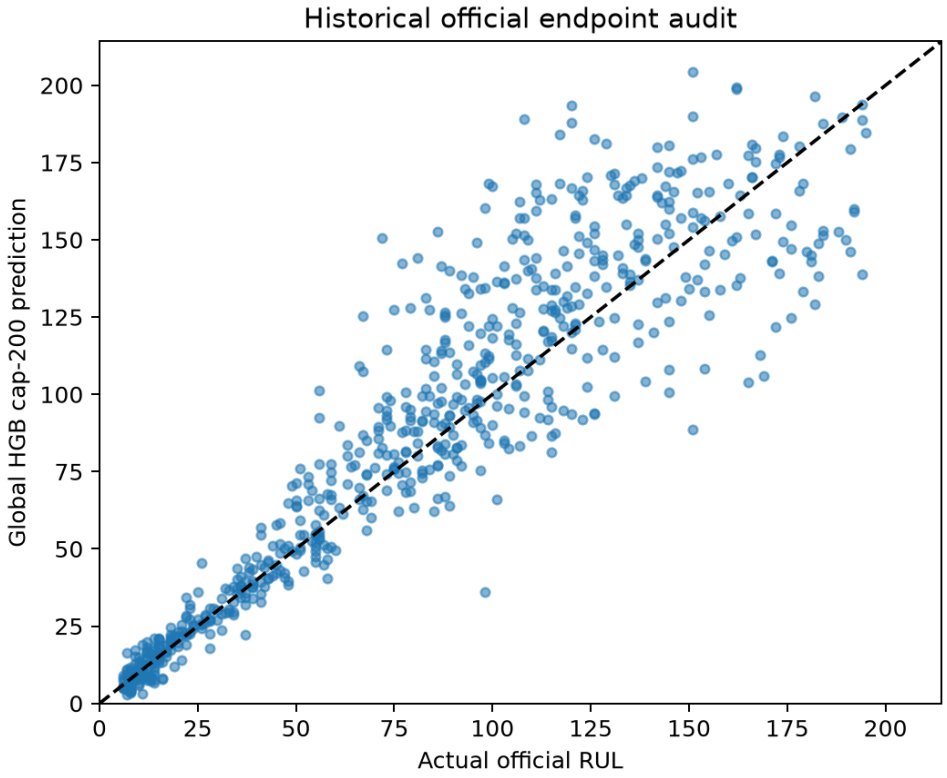
Metrics

RMSE, MAE, R2, PHM08 mean and sum, bias, optimistic prediction rate, and within 5/10/15 cycle rates are saved in the CSV artifacts. Raw target metrics use official RUL without clipping. Capped-target metrics clip both target and prediction to the cap and are diagnostic only.

Cap finding

A cap of 200 allows high-RUL values to be represented and materially helps FD002 and FD004. It changes average bias from underprediction to optimism on FD001 and FD003, so it is not described as universally better.

Subset	Cap100 RMSE	Cap200 RMSE	Cap100 bias	Cap200 bias
FD001	16.292	18.037	-7.688	5.562
FD002	31.128	21.738	-14.767	3.965
FD003	17.273	20.863	-6.686	10.563
FD004	33.391	22.630	-17.451	4.947



Domain-aware experiments

Specialists use a controlled subset identity for benchmarking only. This identity is not assumed to be available for a connected fleet. Each was selected between cap 100 and cap 200 using its own grouped validation.

Subset	Specialist RMSE	Global cap200 RMSE	Specialist PHM08
FD001	18.370	18.037	14.337
FD002	21.713	21.738	25.832
FD003	20.115	20.863	27.958
FD004	24.782	22.630	62.984

Research implementation results

The source-pretrain plus target-residual transfer baseline was tested for FD002/FD004 and rejected. The CPU-bounded compact TCN implemented three dilated residual blocks plus sensor and temporal attention; it was also rejected. The dual-scale Transformer was deliberately not run because the preceding simpler sequence model did not justify more tuning.

Experiment	Subset	Validation RMSE	Validation PHM08
source_pretrain_target_residual_adaptation	FD002	29.342	332.587
source_pretrain_target_residual_adaptation	FD004	39.319	1766957.906
compact_multiscale_tcn_attention_cap200	FD001	35.942	986.137
compact_multiscale_tcn_attention_cap200	FD002	37.911	1221.252
compact_multiscale_tcn_attention_cap200	FD003	52.030	1138285.461
compact_multiscale_tcn_attention_cap200	FD004	48.061	2901941.922
compact_multiscale_tcn_attention_cap200	Pooled	43.615	1180293.687

Critical range and operating workflow

The 0-30 cycle range is reported separately because maintenance decisions become time critical there. The table uses raw official targets and predictions.

Model	Subset	Cases	RMSE	MAE	PHM08	Within 10 cycles
global_hgb_cap100	FD001	25	3.597	2.593	0.294	0.960
global_hgb_cap100	FD002	61	3.262	2.512	0.283	1.000
global_hgb_cap100	FD003	20	1.879	1.654	0.172	1.000
global_hgb_cap100	FD004	53	4.044	3.161	0.394	0.981
global_hgb_cap100	Pooled	159	3.466	2.633	0.308	0.987
global_hgb_cap200	FD001	25	3.433	2.568	0.268	0.960
global_hgb_cap200	FD002	61	4.422	3.079	0.408	0.984
global_hgb_cap200	FD003	20	2.449	1.993	0.229	1.000
global_hgb_cap200	FD004	53	4.728	3.763	0.481	0.962
global_hgb_cap200	Pooled	159	4.190	3.090	0.388	0.975

Competition pipeline

Sensor telemetry -> training-fitted preprocessing -> operating-condition identification -> RUL prediction -> normal-reference reconstruction score -> deterministic triage -> idempotent mock work order -> MLflow. The run used 12000 early-life reference windows, scored 2109 stream windows and emitted 228 work orders.

USP

The system combines a model that handles observable operating-condition changes without subset identity, causal feature controls and a complete explainable maintenance workflow. RUL prediction and anomaly reconstruction are separated, which makes the system honest about the absence of anomaly labels in C-MAPSS.

Recommendation

Use the global cap-200 HGB as the benchmark candidate. Its better pooled result comes with documented FD001/FD003 tradeoffs. Keep the failed specialist, transfer, TCN and routing results in the evidence pack rather than presenting them as improvements. Before live use, calibrate uncertainty, set maintenance thresholds with domain experts and monitor operating-condition drift.